

Mashego Gideon Mabeloane

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EDUCATION

BSc Computer Science, University of the Witwatersrand, Johannesburg
Feb 2023 – November 2025(Completed)

EXPERIENCE

February 2025 - November 2025: Computational and Applied Mathematics Tutor, University of the Witwatersrand -Guide students through university level computational and applied mathematics by providing structured, algorithmic problem-solving support and creating simplified exercises to reinforce key concepts.

- Improved students' understanding of complex mathematical concepts by using divide and conquer approach.
- Strengthened student's analytical and algorithmic thinking by delivering custom exercises.
- Increased students' engagement by creating an environment where they could freely share ideas.

PROJECTS

Freelancio - A dynamic freelance marketplace connecting skilled freelancers with clients. Features include secure payments, real-time messaging, and a robust review system to ensure trust and quality.

- Improved security by implementing authentication using supabase authentication services.
- Ensured seamless and instant messaging between freelancers and clients by leveraging Supabase's real-time services.
- Developed RestAPI for clients to be able to post jobs, accept applications and send contracts

Trailo – Web-Based Hiking Logbook that allows users to track and log hikes, connects friends to hike together and share achievements

- Enhanced hike planning by integrating the OpenWeather API, allowing users to plan trips based on real-time weather information.
- Implemented route-planning, GPS tracking using Google Maps API, enhancing user navigation experience.

Student_classifier – Machine Learning Model that is used in StuddyBuddy AI web application to improves students' academic performance

- Developed a **Gradient Boosting model** to classify students based on exam scores and attendance.
- Performed **data preprocessing, feature engineering, and model evaluation** to optimize accuracy.
- Exposed the model via a **REST API** using **FASTAPI** and integrated it into **StuddyBuddy**, enabling **real-time, data-driven academic insights** for students and educators.

StudyBuddy – full-stack web application that helps students track study activities, collaborate with peers, and receive personalized academic support using data-driven insights

- Integrated **Gradient Boosting model** to classify students based on data, enabling personalized study recommendations.
- Planning to integrate LLMs to summarise or explain more of the lecture notes and generate quizzes depending on what type of the student
- Planning integration of the **YouTube API** and curated educational resources to enhance concept understanding
- Designing a **peer collaboration system**, including friend connections and real-time chat functionality.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, JavaScript, Dart

Frameworks/Tools: React +Vite, Flutter/Dart, REST APIs, FastAPI

AI and Machine Learning: **Supervised** and Unsupervised Learning, Model Evaluation and Feature Engineering, Classification, Regression, Clustering

Data Handling and Analysis: Data Cleaning and Preprocessing, Exploratory Data Analysis (EDA)

Database and Backend Services: Supabase, Firebase, MySQL